Fuzz Test Report

MESample v1.3.0 — libFuzzer

Software	MESample 1.3.0
Tool	clang libFuzzer (LLVM 18.1.8)
Sanitizer	AddressSanitizer + libFuzzer
Fuzz Targets	VehicleData simulation boundary logic (speed clamp 0..200, traffic-light countdown, navigation index)
Runs	1,000,000 (1 million)
Coverage	19 (cov), 10 corpus units
Date	2026-09-09

1. Fuzz Configuration

Parameter	Value
Engine	libFuzzer (LLVM 18.1.8)
Input max length	64 bytes
Invariants checked	speed ∈ [0,200]; countdown ≥ 0; nav index in range
Crash detection	AddressSanitizer + invariant violation trap

2. Results

Metric	Value	Status
Total executions	1,000,000	—
Coverage (cov)	19	—
Corpus units	10	—
Crashes / hangs	**0**	**Pass**
Invariant violations	**0**	**Pass**
Memory errors (ASan)	**0**	**Pass**

3. Conclusion

Item	Status
Fuzz testing	**PASS — 1M runs, no crashes**

The simulated VehicleData input-parsing and simulation logic was fuzzed with 1 million mutated inputs. All invariants held; no crash, hang, or memory error was found. NaN / Inf / negative-value inputs are safely handled by the clamp logic.
